

# FinWallet Fail Path - Failure and Recovery Scenarios

This document describes the important FinWallet v1 failure behaviors. It does not stop at HTTP status codes; it also explains what happens to customer balances, Ledger, idempotency and retry state.

## 0. Core safety rule

Normal customer traffic goes through YARP Gateway.

```
{{gateway}} = http://localhost:8080
```

General financial rules:

- No external HTTP call runs inside an open MSSQL financial transaction.
- A Completed money movement is never rolled back because communication failed.
- Transfer/Purchase fail closed when fraud is uncertain.
- MSSQL is the financial source of truth.

## 1. FAIL - Missing or invalid JWT

```
GET {{gateway}}/api/v1/wallets
```

Gateway rejects the request before forwarding it.

```
{
  "isSuccess": false,
  "code": "GATEWAY_UNAUTHORIZED",
  "message": "A valid access token is required by the gateway.",
  "data": null,
  "errors": []
}
```

HTTP: 401 Unauthorized.

- MSSQL change: none
- Wallet/Ledger change: none
- Retry: after obtaining a valid JWT

## 2. FAIL - Registration country/phone mismatch

Example: TR is selected but an incompatible phone prefix is supplied.

```
POST {{gateway}}/api/v1/auth/register
Content-Type: application/json
```

```
{
  "countryCode": "TR",
  "phoneNumber": "+994501234567",
  "email": "invalid@example.test",
  "password": "Example-Password-123!"
}
```

Response:

```
{
  "isSuccess": false,
  "code": "REGISTRATION_NOT_ALLOWED",
  "message": "The registration country or phone number is not eligible.",
  "data": null,
  "errors": []
}
```

HTTP: 400 Bad Request.

## 3. FAIL - OTP invalid, expired or consumed

```
POST {{gateway}}/api/v1/auth/registration/verify
Content-Type: application/json
```

```
{
  "customerId": "11111111-1111-1111-1111-111111111111",
  "code": "999999"
}
```

Response:

```
{
  "isSuccess": false,
  "code": "INVALID_REGISTRATION_OTP",
  "message": "The verification code is invalid, expired or already consumed.",
  "data": null,
  "errors": []
}
```

HTTP: 400 Bad Request.

Resending too quickly returns:

429 OTP\_RESEND\_RATE\_LIMIT

## 4. FAIL - Invalid login or logout

Invalid credentials:

```
{
  "isSuccess": false,
  "code": "INVALID_CREDENTIALS",
  "message": "The supplied credentials are invalid.",
  "data": null,
  "errors": []
}
```

HTTP: 401 Unauthorized.

After the failed-login limit:

```
{
  "isSuccess": false,
  "code": "AUTH_TEMPORARILY_LOCKED",
  "message": "Authentication is temporarily unavailable for this credential.",
  "data": null,
  "errors": []
}
```

HTTP: 429 Too Many Requests.

## 5. FAIL - FakeBank unavailable during BankAccount opening

Client request:

```
POST {{gateway}}/api/v1/bank-accounts
Authorization: Bearer {{token}}
Content-Type: application/json
```

```
{
  "walletId": "44444444-4444-4444-4444-444444444444"
}
```

FinWallet first durably writes the internal BankAccount in Opening state, closes the SQL transaction, then calls FakeBank.

Example provider errors:

```
FAKE_BANK_UNAVAILABLE
BANK_PROVIDER_TIMEOUT
BANK_PROVIDER_NETWORK_ERROR
```

A retryable provider exception can be exposed as 503 Service Unavailable. The durable internal BankAccount is not lost. A deterministic provider request key is derived from the same internal ID, so retries should not open duplicate external accounts.

```
Wallet balance change: none
Ledger change: none
```

BankAccount durable state: may remain Opening  
Retry: safe with same provider request key

## 6. RESILIENCE - BankDeposit provider timeout/network

Client:

```
POST {{gateway}}/api/v1/bank-movements/deposits
Authorization: Bearer {{token}}
Idempotency-Key: fail-bank-deposit-001
Content-Type: application/json

{
  "bankAccountId": "55555555-5555-5555-5555-555555555555",
  "amount": 1000.00
}
```

FinWallet first creates durable bank-movement and idempotency state. If the FakeBank call times out or has a network failure, the handler does not finalize the operation as a financial failure.

Expected lifecycle response:

```
{
  "isSuccess": true,
  "code": "BANKDEPOSIT_PENDING",
  "message": "BankDeposit state is Pending.",
  "data": {
    "operation": "BankDeposit",
    "amount": 1000.00,
    "currency": "TRY",
    "state": "Pending",
    "wasReplay": false
  },
  "errors": []
}
```

HTTP: 202 Accepted.

BankMoneyMovementBackgroundService later retries/polls the due operation.

```
Wallet balance change: not yet
Ledger posting: not yet
Durable transaction: Processing/Pending
Retry: background worker
```

## 7. FAIL - External FakeBank account has insufficient money

The client requests 1,000 TRY BankDeposit while the external account contains only 100 TRY.

FakeBank prevents negative provider balance and produces a transaction conflict/terminal error.

Typical provider code:

```
BANK_TRANSACTION_CONFLICT
```

FinWallet converts the non-retryable provider error into terminal Failed state.

The public bank-movement result may represent this terminal business state as:

```
{
  "isSuccess": true,
  "code": "BANKDEPOSIT_FAILED",
  "message": "BankDeposit state is Failed.",
  "data": {
    "operation": "BankDeposit",
    "amount": 1000.00,
    "currency": "TRY",
    "state": "Failed",
    "wasReplay": false
  },
  "errors": []
}
```

```
External bank balance: unchanged
Wallet available balance: unchanged
```

Ledger posting: none  
Idempotency: terminal failure state

## 8. FAIL - Insufficient Wallet balance for transfer

POST {{gateway}}/api/v1/transfers  
Authorization: Bearer {{token}}  
Idempotency-Key: fail-transfer-balance-001  
Content-Type: application/json

```
{
  "sourceWalletId": "11111111-aaaa-aaaa-aaaa-111111111111",
  "destinationWalletId": "22222222-bbbb-bbbb-bbbb-222222222222",
  "amount": 5000.00
}
```

If source available balance is 500 TRY:

```
{
  "isSuccess": false,
  "code": "INSUFFICIENT_BALANCE",
  "message": "The source wallet has insufficient available balance.",
  "data": null,
  "errors": []
}
```

HTTP: 409 Conflict.

The atomic posting transaction rolls back; no partial balance or Ledger change remains.

## 9. FAIL - Fraud provider unavailable

For Transfer/Purchase, external FakeFraud is mandatory unless internal fraud already denied the operation. Network/timeout/invalid response causes fail-closed behavior:

```
{
  "isSuccess": false,
  "code": "FRAUD_DEPENDENCY_UNAVAILABLE",
  "message": "The required fraud service is temporarily unavailable.",
  "data": null,
  "errors": []
}
```

HTTP: 503 Service Unavailable.

Wallet balance: unchanged  
Ledger: unchanged  
Financial posting: never starts

## 10. FAIL - Same Idempotency-Key reused with different request

First request:

```
{
  "destinationWalletId": "22222222-bbbb-bbbb-bbbb-222222222222",
  "amount": 100.00
}
```

Second request with the same key:

```
{
  "destinationWalletId": "33333333-cccc-cccc-cccc-333333333333",
  "amount": 200.00
}
```

Response:

```
{
  "isSuccess": false,
  "code": "IDEMPOTENCY_CONFLICT",
  "message": "The Idempotency-Key was already used with a different transfer request.",
  "data": null,
  "errors": []
}
```

HTTP: 409 Conflict.

This prevents duplicate protection from replaying the wrong request.

## 11. FAIL - Notification provider down after money completed

Assume BankDeposit or Purchase already committed money and Outbox in the same transaction. The FinancialTransaction is now Completed.

If FakeCommunication returns 503/timeout:

```
FinancialTransaction: remains Completed
Wallet/Ledger: not rolled back
Outbox: not Processed
Outbox: rescheduled with bounded backoff
```

Worker failure codes include:

```
COMMUNICATION_UNAVAILABLE
OUTBOX_DISPATCH_ERROR
OUTBOX_RECIPIENT_UNAVAILABLE
OUTBOX_INVALID_PAYLOAD
```

The previously returned financial success is never reversed because communication failed.

## 12. FAIL - Rate limit / method / content type

Rate limit:

```
{
  "isSuccess": false,
  "code": "RATE_LIMITED",
  "message": "Too many requests.",
  "data": null,
  "errors": []
}
```

TRACE or CONNECT:

```
405 METHOD_NOT_ALLOWED
```

POST/PUT/PATCH with a non-JSON body:

```
415 UNSUPPORTED_MEDIA_TYPE
```

## 13. Failure behavior summary

| Failure                               | Money changes?    | Ledger written? | Retry                     |
|---------------------------------------|-------------------|-----------------|---------------------------|
| Gateway unauthorized                  | No                | No              | With new JWT              |
| Invalid OTP/login                     | No                | No              | Per security policy       |
| Bank-account provider timeout         | No                | No              | Same request key          |
| BankDeposit transient provider error  | Not yet           | Not yet         | Background                |
| External-bank insufficient balance    | No                | No              | Terminal failure          |
| Transfer insufficient balance         | No                | No              | New request after funding |
| Fraud unavailable                     | No                | No              | After dependency recovery |
| Idempotency conflict                  | No                | No              | Correct new key/request   |
| Notification unavailable after commit | Already completed | Yes             | Outbox retry              |